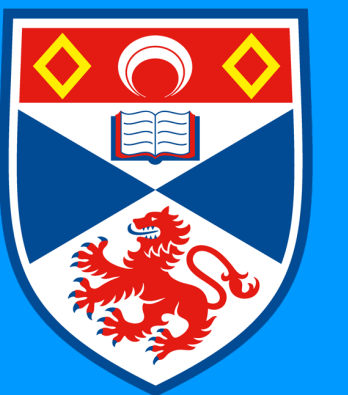


Visualising the Abraham-Minkowski Light Field Momentum

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Motivation

Comets' tails point away from the Sun due to radiation pressure, i.e. light – electric and magnetic fields – carries momentum which can be transferred to matter. In vacuum, the electromagnetic momentum density is unambiguously

$$\mathbf{g} = \epsilon_0(\mathbf{E} \times \mathbf{B})$$

However, what happens when light travels in a transparent medium? Matter is subject to polarisation and magnetisation, so the effective field momentum is modified [1]. In the early 1900s, Abraham [2] and Minkowski [3] developed two opposing expressions based on classical electrodynamics. More than a century later, multiple theoretical and experimental attempts [4] have been made to resolve the controversy.

The goal of this project is to produce computational tools to visualise electromagnetic fields and their momenta as they propagate through media.

Theoretical Background

Plane Waves

They are solutions to Maxwell's equations in free space:

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad \mathbf{B} = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

where $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields, $\mathbf{E}_0$ and $\mathbf{B}_0$ are the amplitude vectors (pointing perpendicular to $\mathbf{k}$ and each other), $\mathbf{k}$ is the wavevector, ω is the angular frequency, $\mathbf{r}$ is the position vector and t is time.

Maxwell Stress Tensor

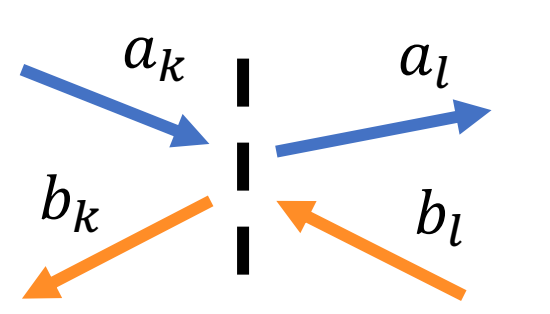
It is related to the conservation of linear and angular momentum, and is at the root of electromagnetic forces:

$$\vec{\sigma} = \epsilon_0 E_i E_j + \frac{1}{\mu_0} B_i B_j - \frac{1}{2} \delta_{ij} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) \rightarrow \nabla \cdot \vec{\sigma} = \mathbf{f} + \frac{1}{c^2} \frac{\partial \mathbf{S}}{\partial t}$$

where $i, j = x, y, z$ correspond to the projections of the field onto the Cartesian directions, μ_0 and ϵ_0 are respectively the vacuum permeability and permittivity, $\mathbf{f}$ is the force density and $\mathbf{S}$ is the Poynting vector. In free space, $\mathbf{S} = \mathbf{g}c^2$.

Scattering (S) & Transfer (M) Matrices

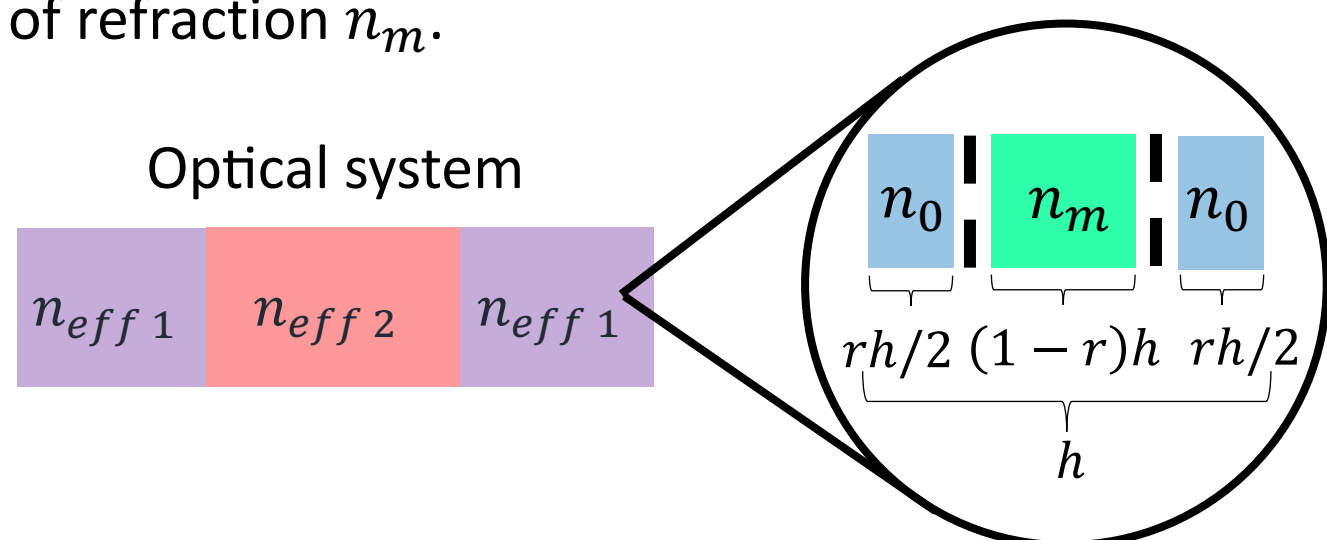
They are tools for calculating field propagation.


$$\vec{S}_{kl} = \begin{pmatrix} r_{kl} & t_{lk} \\ t_{kl} & r_{lk} \end{pmatrix} \quad \begin{pmatrix} b_k \\ a_l \end{pmatrix} = \vec{S}_{kl} \begin{pmatrix} a_k \\ b_l \end{pmatrix}$$
$$\vec{M}_{kl} = \begin{pmatrix} m_{kk} & m_{kl} \\ m_{lk} & m_{ll} \end{pmatrix} \quad \begin{pmatrix} a_l \\ b_l \end{pmatrix} = \vec{M}_{kl} \begin{pmatrix} a_k \\ b_k \end{pmatrix}$$

where r_{kl} and t_{kl} are the reflection and transmission coefficients going from site k to site l with respect to an interface, m_{kl} can be straightforwardly obtained through combinations of r_{kl} and t_{kl} , and a_k and b_k are respectively the complex amplitudes for the forward and backward propagating modes of the field at k .

Periodic Crystal Model

Periodic structure: vacuum, interface, medium, interface, vacuum. Period width h , vacuum portion r , vacuum index of refraction n_0 , and medium index of refraction n_m .



The effective refractive index of a crystal n_{eff} depends on r and n_m , and is calculated from the phase shift induced by its periods.

Numerical Implementation

Input

- Incidence wavevectors
- Optical system properties (period number, width, refraction indices, etc.)

Create Simulation

- Concatenated M-matrices of the composite optical system
- Element-by-element or period-by-period

Fields

- Generate output with the eigenvectors of the exit crystal
- Backpropagate through the system to produce fields between periods
- Apply spectral and temporal modulation

Momentum Flux and Force Density

- Compute the MST for positions, times and wavenumbers
- Project onto z-direction and take the derivative

Results

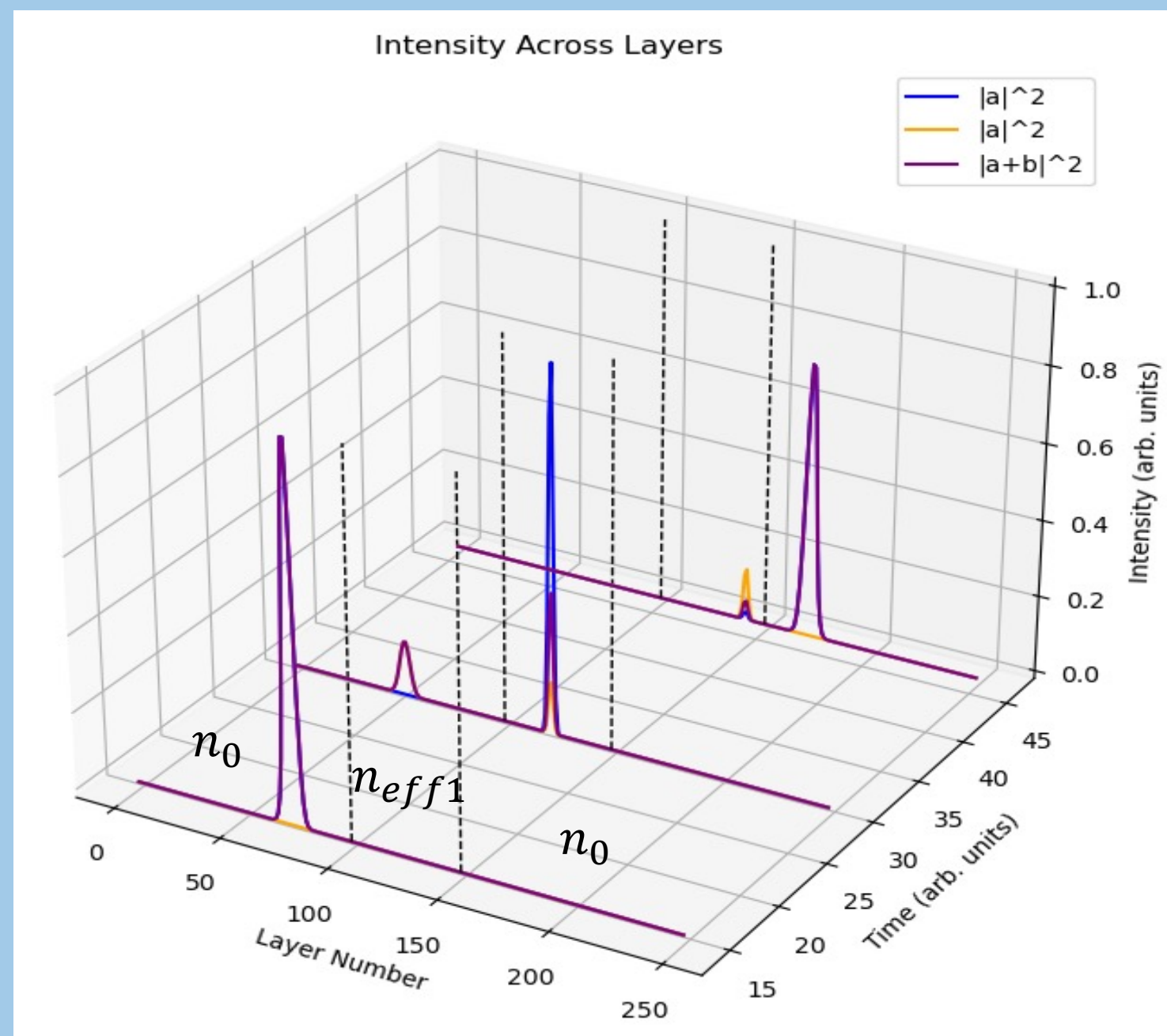


Figure 1: Field intensities at different positions in the optical system at 3 different points in time for a pulse travelling from vacuum to a medium with n_{eff1} to vacuum again.

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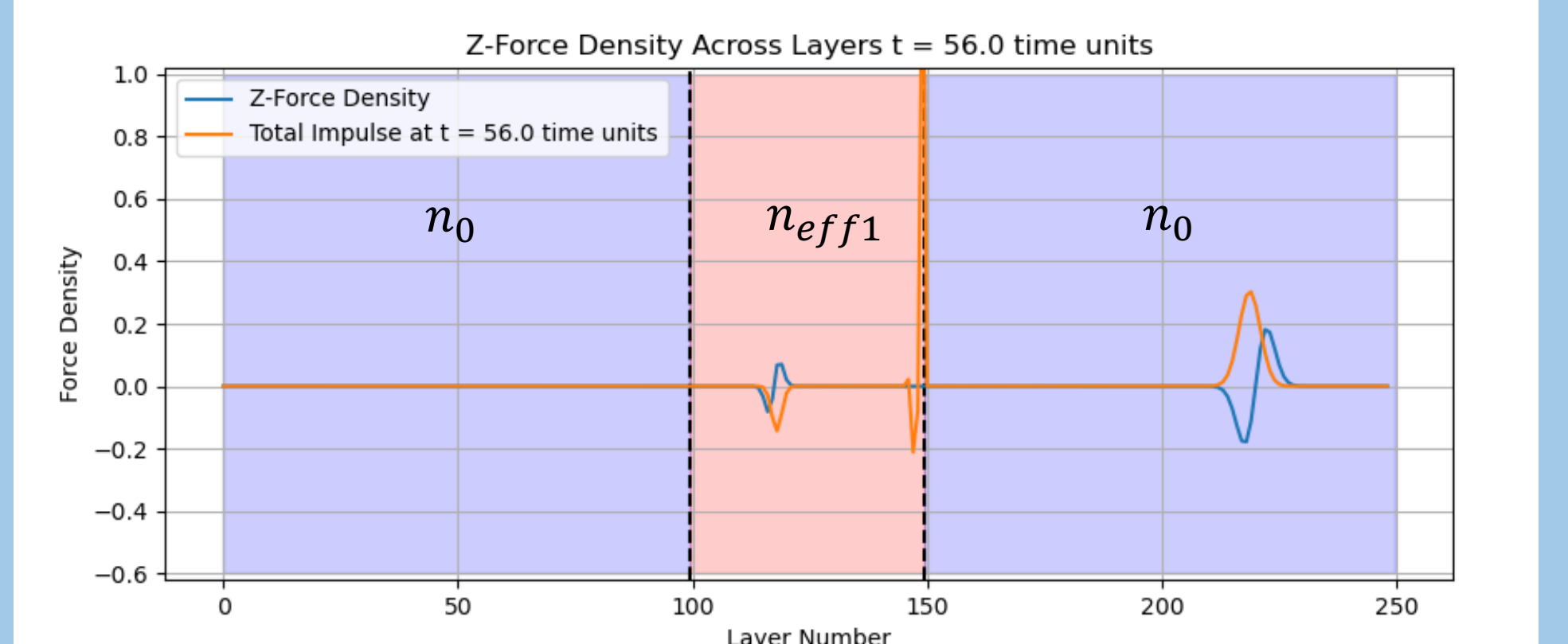


Figure 2: The force density in the z-direction and the total impulse density at time $t=56.0$ (arb.units) for a pulse travelling from vacuum to a medium with n_{eff1} to vacuum again.

Conclusion & Discussion

We have developed a computational approach compatible with both Abraham (counter-propagating field $\rightarrow$ momentum reduced) and Minkowski (momentum flux peak enhancement) momentum transfers.

Moving forward, we aim to:

- Quantify the effect (peak reduction and increase) and compare it with the crystal effective index of refraction.
- Calculate the momentum transfer using the Lorentz force acting on the induced surface currents and charges at the interfaces surrounding the medium part of the periods.
- Test the dependence of the momentum transfer with respect to the angle of incidence and the period width.

References

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